Aplicação da Lógica Fuzzy para o Controlador de Irrigação

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Disponível em:

https://github.com/Ashley-J-George/Sprinkle_Irrigation_Controller/blob/main/Sprinkle_Irrigation_Controller.ipynb

Adaptado por: Dr. Arnaldo de Carvalho Junior – 2025

https://colab.research.google.com/drive/1f0TzUnk9rSX1mvnculBrv2lAEwxHMrNy?usp=sharing

Controlador do Sprinkle de Irrigação

Aplicação da lógica difusa para o controlador de aspersão na irrigação por gotejamento

Controlar quanto tempo um aspersor deve funcionar, dada a temperatura e a umidade atuais

Entradas:

- Temperatura: em grau Celsius
- Umidade: em porcentagem (%)
- Saída: Duração do sprinkler em minutos

```
# Instalando a biblioteca scikit-fuzzy
%%shell
!pip3 install scikit-fuzzy

# Importando Bibliotecas
HUMIDITY = 'humidity'
SPRINKLER_DURATION = 'sprinkler_duration'
TEMPERATURE = 'temperature'

# Declarando Constantes
# Temperature's fuzzy linguistics
COLD = 'Cold'
COOL = 'Cool'
NORMAL = 'Normal'
WARM = 'Warm'
```

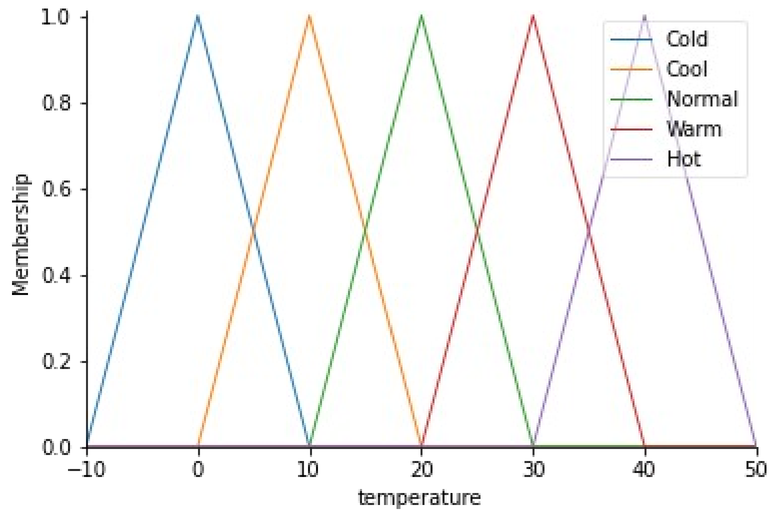
```
HOT = 'Hot'

# Humidity's fuzzy linguistics
DRY = 'Dry'
MOIST = 'Moist'
WET = 'Wet' # Sprinkler duration's fuzzy linguistics
SHORT = 'Short'
MEDIUM = 'Medium'
LONG = 'Long'

# Ajustando os antecedentes e consequentes
temperature = ctrl.Antecedent(np.arange(-10,55,5), TEMPERATURE)
humidity = ctrl.Antecedent(np.arange(0,105,5), HUMIDITY)
sprinkler_duration = ctrl.Consequent(np.arange(0,105,5),
SPRINKLER_DURATION)
# Assinalando valores membros para Temperatura usando função membro
Triangular
cold_parameter = [-10,0,10]
cool_parameter = [0,10,20]
normal_parameter = [10,20,30]
warm_parameter = [20,30,40]
hot_parameter = [30,40,50]

temperature[COLD] = fuzz.trimf(temperature.universe, cold_parameter)
temperature[COOL] = fuzz.trimf(temperature.universe, cool_parameter)
temperature[NORMAL] = fuzz.trimf(temperature.universe,
normal_parameter)
temperature[WARM] = fuzz.trimf(temperature.universe, warm_parameter)
temperature[HOT] = fuzz.trimf(temperature.universe, hot_parameter)
temperature.view()

# Assinalando valores membros para Umidade usando função membro
Triangular
dry_parameter = [0,25,50]
moist_parameter = [25,50,75]
wet_parameter = [50,75,100]

humidity[DRY]    =    fuzz.trimf(humidity.universe,    dry_parameter)
humidity[MOIST]   =   fuzz.trimf(humidity.universe,   moist_parameter)
humidity[WET]    =    fuzz.trimf(humidity.universe,    wet_parameter)
humidity.view()
```

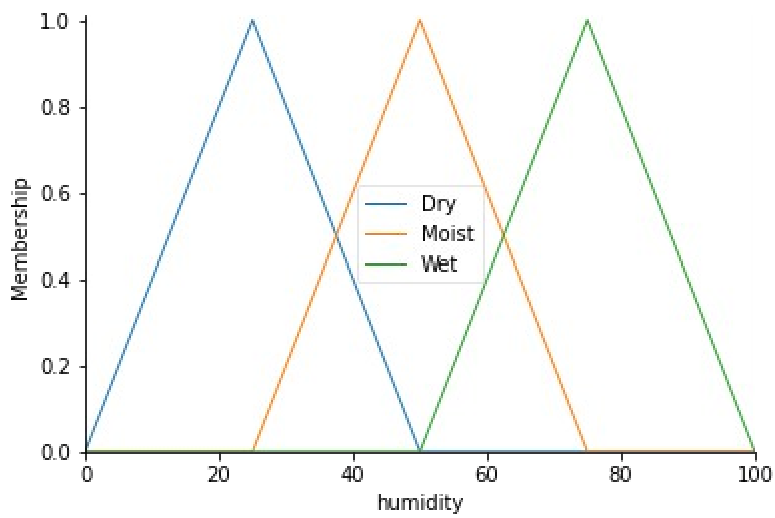
```python
# Atribuindo valores dmembro à duração do sprinkler usando a função membro triangular
short_parameter = [0,25,50]
medium_parameter = [25,50,75]
long_parameter = [50,75,100]

sprinkler_duration[SHORT]  =  fuzz.trimf(sprinkler_duration.universe, short_parameter)
sprinkler_duration[MEDIUM]  =  fuzz.trimf(sprinkler_duration.universe, medium_parameter)
sprinkler_duration[LONG]   =   fuzz.trimf(sprinkler_duration.universe, long_parameter) sprinkler_duration.view()

# Configurando as Regras de Performance
rule1 = ctrl.Rule(humidity[DRY] & temperature[COLD], sprinkler_duration[SHORT])
rule2 = ctrl.Rule(humidity[DRY] & temperature[COOL], sprinkler_duration[SHORT])
rule3 = ctrl.Rule(humidity[DRY] & temperature[NORMAL], sprinkler_duration[MEDIUM])
rule4 = ctrl.Rule(humidity[DRY] & temperature[WARM], sprinkler_duration[LONG])
rule5 = ctrl.Rule(humidity[DRY] & temperature[HOT], sprinkler_duration[LONG])
rule6 = ctrl.Rule(humidity[MOIST] & temperature[COLD], sprinkler_duration[SHORT])
rule7 = ctrl.Rule(humidity[MOIST] & temperature[COOL], sprinkler_duration[SHORT])
rule8 = ctrl.Rule(humidity[MOIST] & temperature[NORMAL], sprinkler_duration[MEDIUM])
rule9 = ctrl.Rule(humidity[MOIST] & temperature[WARM], sprinkler_duration[MEDIUM])
rule10 = ctrl.Rule(humidity[MOIST] & temperature[HOT], sprinkler_duration[LONG])
rule11 = ctrl.Rule(humidity[WET] & temperature[COLD], sprinkler_duration[SHORT])
rule12 = ctrl.Rule(humidity[WET] & temperature[COOL], sprinkler_duration[SHORT])
rule13 = ctrl.Rule(humidity[WET] & temperature[NORMAL], sprinkler_duration[SHORT])
```

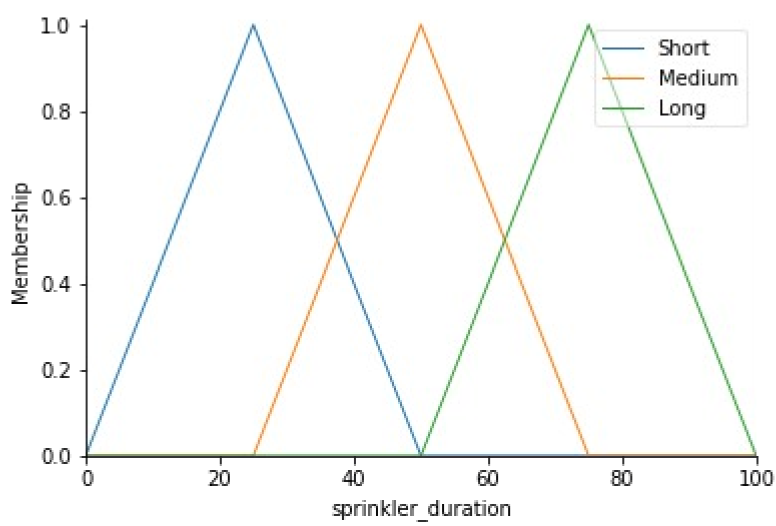
```
rule14 = ctrl.Rule(humidity[WET] & temperature[WARM],
sprinkler_duration[MEDIUM])
rule15 = ctrl.Rule(humidity[WET] & temperature[HOT],
sprinkler_duration[LONG])

# Lista de Regras
rule_list = [rule1, rule2, rule3, rule4, rule5, rule6, rule7, rule8,
rule9, rule10, rule11, rule12, rule13, rule14, rule15]

# Invocando o sistema com as regras Fuzzy
sprinkler_ctrl = ctrl.ControlSystem(rule_list)
sprinkler_analysis = ctrl.ControlSystemSimulation(sprinkler_ctrl)

# Avaliando o resuldado do sistema
sprinkler_analysis.input[TEMPRATURE] = 45
sprinkler_analysis.input[HUMIDITY] = 30

sprinkler_analysis.compute() print(f'Evaluated Result:
{str(round(sprinkler_analysis.output[SPRINKLER_DURATION], 2))} Min')
# Visualizando o resultado
sprinkler_duration.view(sim=sprinkler_analysis)
```

Regras

H \ T	Cold	Cool	Normal	Warm	Hot
Dry	S	S	M	L	L
Moist	S	S	M	M	L
Wet	S	S	S	M	L


1.0
0.8
0.6
0.4
0.2
0.0
Membership
−10
0
10
20
30
40
50
temperature
Cold
Cool
Normal
Warm
Hot


1.0
0.8
0.6
0.4
0.2
0.0
Membership
0
20
40
60
80
100
humidity
Dry
Moist
Wet


1.0
0.8
0.6
0.4
0.2
0.0
Membership
0
20
40
60
80
100
sprinkler_duration
Short
Medium
Long

Resultados:
Evaluated Result: 75.0 Min